

# PHARMACY DEMAND INTELLIGENCE

*Synthetic Demonstration / Sales • Demand • Availability • Ordering • Stock*

## Project Details

A synthetic pharmacy operations analytics demonstration exploring how sales, customer requests, product availability, stockouts and ordering patterns can be combined to support better operational decisions.

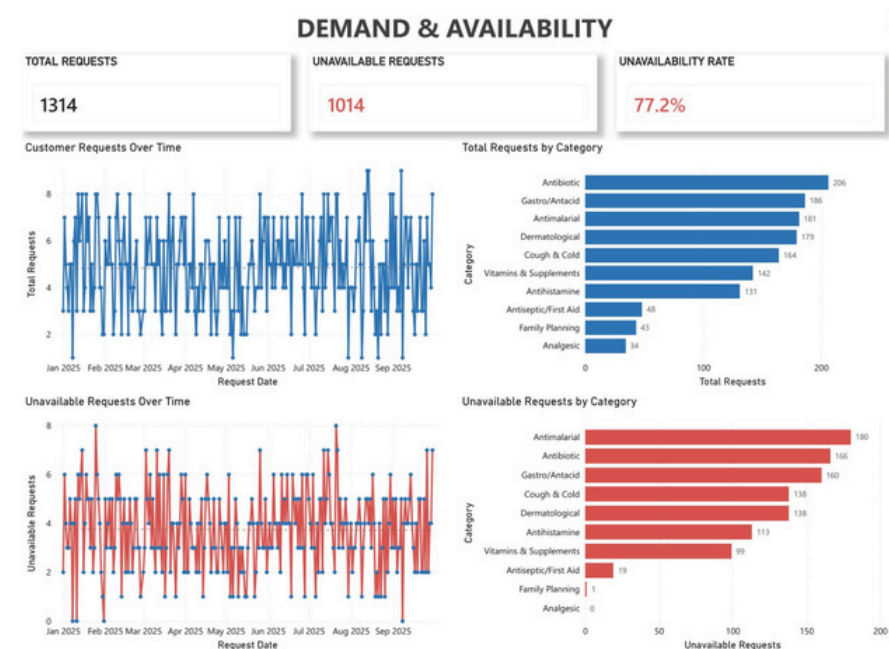
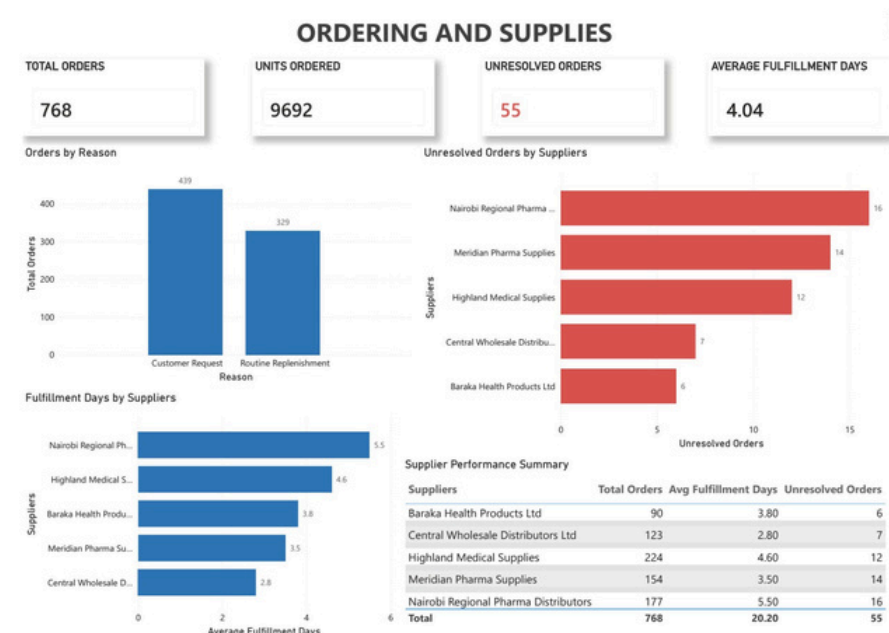
## Project Purpose

The demonstration is designed to answer:

- What products are actually selling?
- What products are customers requesting?
- Which requests are not being fulfilled?
- Where do stock availability problems appear?
- How do ordering patterns and supplier fulfillment affect operations?

**Data:** Synthetic / Period: Jan–Sep 2025

**Tools:** Python • MySQL • Power BI



Power BI

# FROM RAW OPERATIONAL DATA TO DECISION SUPPORT

## ***Analytics Workflow***

***Synthetic Data***



***Data Generation & QC***



***MySQL Database***



***Analytical SQL Views***



***Power BI Model***



***Decision Insights***

***Key principle:*** Sales alone do not represent total demand. Combining sales, requests and availability provides a more complete view of pharmacy operations.

## ***Sales***

- *Revenue*
- *Units sold*
- *Product/category performance*

## ***Customer Demand***

- *Customer requests*
- *Requested products/categories*
- *Demand patterns*

## ***Availability***

- *Unavailable requests*
- *Unavailability rate*
- *Stockout days*

## ***Ordering***

- *Orders placed*
- *Units ordered*
- *Order reasons*

## ***Suppliers***

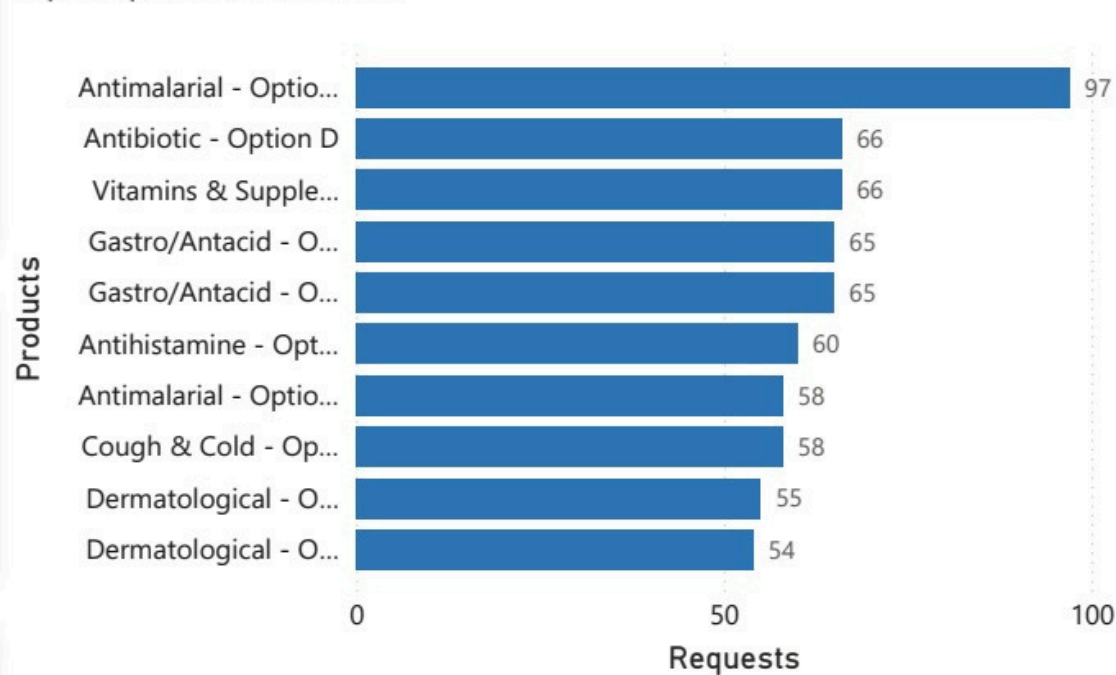
- *Fulfillment time*
- *Unresolved orders*
- *Supplier patterns*

## ***Assortment***

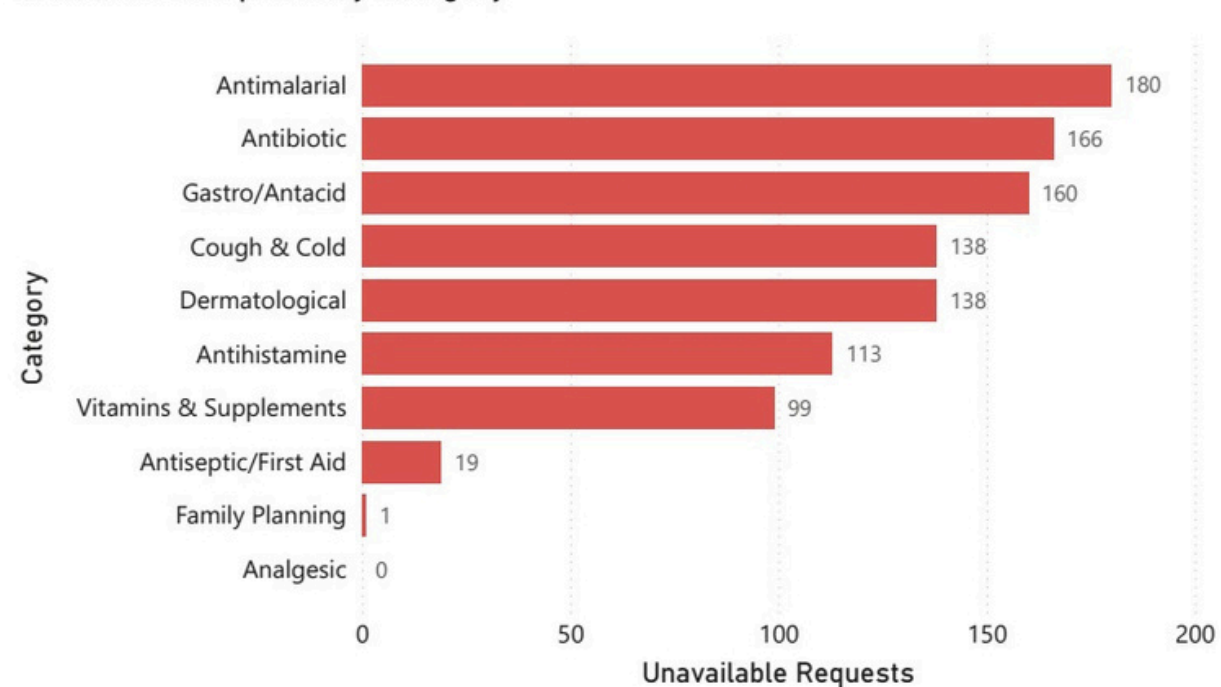
- *Routinely stocked products*
- *Frequently requested products*
- *Products requiring review*

# CUSTOMER DEMAND & AVAILABILITY

Top Requested Products



Unavailable Requests by Category



*Antimalarial – Option C*

*97 requests · Not routinely stocked*

*Highest request volume among products reviewed*

## **What the data reveals**

Customer requests reveal demand that sales data alone cannot capture.

## **Important interpretation**

This does not mean the product should automatically be stocked. It identifies a product for further review based on observed demand and availability patterns.

*Demand intelligence connects customer requests with availability gaps to identify where service may be constrained.*

# Stock, Ordering & Supplier Performance

## Stock

Stockout Days: 23

Total Units Sold: 8,756

*-Stock analysis helps identify products experiencing availability pressure despite recorded sales.*

## Ordering

Total Orders: 768

Units Ordered: 9,692

Unresolved Orders: 55

*-Orders can be examined by reason to distinguish routine replenishment from customer-driven or shortage-driven ordering.*

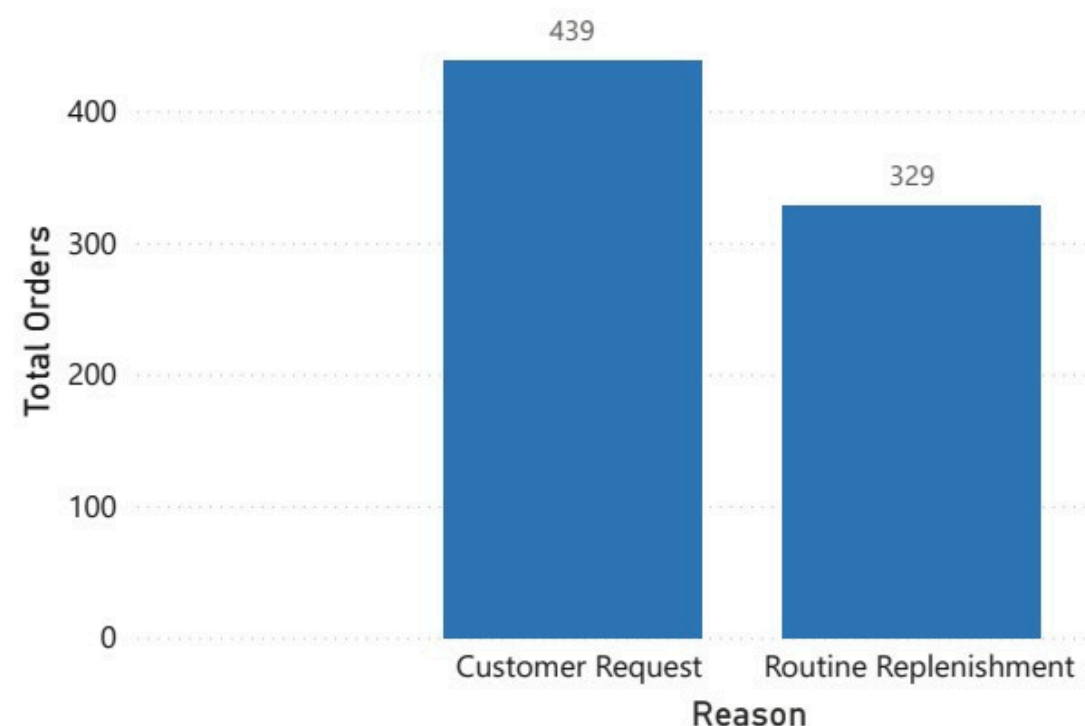
*Inventory availability is influenced by both internal demand patterns and the speed and reliability of replenishment.*

## Supplier Performance

Average Fulfillment: 4.04 days

*-Supplier analysis adds an upstream operational dimension by showing how fulfillment patterns may affect inventory availability.*

Orders by Reason



# Key Findings & What This Demonstrates

## Analytical Workflow Demonstrated

### **Python/Pandas**

*-Synthetic data generation & quality checks*

### **MySQL**

*-Relational database, validation & analytical views*

### **Power BI**

*-Interactive dashboard & decision-support visualization*

### **Demand ≠ Sales**

*-Customer requests reveal demand that may not appear in completed sales.*

### **Availability Matters**

*-1,014 requests were unavailable, highlighting a substantial service-availability signal in the synthetic dataset.*

### **Products for Review**

*-High-request products that are not routinely stocked can be flagged for further operational review.*

### **Ordering Has Context**

*-Order reasons, fulfillment time and unresolved orders provide additional insight beyond sales performance.*

*This is a synthetic demonstration, not a live pharmacy analysis. Its purpose is to demonstrate how operational data can be transformed into practical decision-support insights.*

